FAST- National University of Computer and Emerging Sciences, Karachi.

FAST School of Computing, Fall 2025

CS1005-Discrete Structures

Assignment # 3

Instructions: **Maximum Points: 100**

1- This is a handwritten assignment. You must submit the hard copy in your class on December 5th, 2025, along with the scanned copy on Google Classroom.
2- Write the question number instead of writing the whole question.
3- You can only use A4 size paper to solve the assignment.
4- You must write your student ID and section at the top and your signature at the bottom of each page.

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1. (a) Prove: For all integers *a*, *b*, and *c*,
   (i) If $a|b$ and $a|c$, then $a|(b + c)$. (ii) If $a|b$, then $a|(bc)$. (iii) If $a|b$ and $b|c$, then $a|c$.
   (b) Prove or disprove that the product of two consecutive integers is an even integer.
   (c) Prove or disprove that the product of two odd integers and an even integer is an even integer.

2. (a) Prove the statement: There are real numbers a and b such that $\sqrt{(a+b)} = \sqrt{a} + \sqrt{b}$.
   (b) Prove that the square of any rational number is a rational number.
   (c) Prove the statement: There is an integer n > 5 such that $2^n - 1$ is prime.

3. (a) Prove by contradiction method, the statement: If n and m are odd integers, then n + m is an even integer.
   (b) Prove by Contradiction that for any integer a and any prime number p, if p | a, $P \nmid (a + 1)$.
   (c) Prove by Contradiction that the set of prime numbers is infinite.

4. (a) Prove by contraposition that if |x| > 1 then x > 1 or x < –1 for all x € R.
   (b) Prove by contraposition: For all integers m and n, if m + n is even then m and n are both even or m and n are both odd.
   (c) Prove by contraposition that if *ab* is divisible by 4, then *a* and *b* are positive, even integers.

5. (a) Find a counter example to the proposition:
   (i) For every prime number n, n + 2 is prime. (ii) Every even number is divisible by 4.
   (b) Find a counter example:
   (i) Every positive integer is a prime number.
   (ii) The square of a real number is always greater than the number itself.
   (c) Prove or disprove that the product of any two irrational numbers is an irrational number.

6. (a) Prove by contradiction that $6 - 7\sqrt{2}$ is irrational.
   (b) Prove by contradiction that $\sqrt{2} + \sqrt{3}$ is irrational.
   (c) Proof that for all integers z, $4(z^2 + z + 1) - 3z^2$ is a perfect square.

7. (a) Prove by mathematical Induction. $P(n) = 1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for all natural numbers *n*.
   (b) The sum of the cube of the first n natural numbers is $\frac{1}{4}n^2(n + 1)^2$.
   (c) Prove by mathematical induction: $\frac{1}{1.2} + \frac{1}{2.3} + \cdots \frac{1}{n(n+1)} = \frac{n}{n+1}$ for all integers $n \geq 1$.

8. (a) What are the quotient and remainder when:
   (i) 19 is divided by 7? (ii) −111 is divided by 11? (iii) 789 is divided by 23?
   (iv) 1001 is divided by 13? (v) 10 is divided by 19? (vi) 3 is divided by 5?
   (vii) −1 is divided by 3? (viii) 4 is divided by 1?
   (b) Find a div m and a mod m when
   i) a = −111, m = 99. ii) a = −9999, m = 101. iii) a = 10299, m = 999. iv) a = 123456, m = 1001.
   (c) Decide whether each of these integers is congruent to 5 modulo 17.
   (i) 80 (ii) 103 (iii) −29 (iv) −122

9. (a) Determine whether the integers in each of these sets are pairwise relatively prime.
   (i) 11, 15, 19 (ii) 14, 15, 21 (iii) 12, 17, 31, 37 (iv) 7, 8, 9, 11
   (b) Find the prime factorization of each of these integers.
   (i) 88 (ii) 126 (iii) 729 (iv) 1001 (v) 1111 (vi) 909

(c) Find the GCD and LCM of the following using prime factors.
(i) 11, 15, 105 (ii) 14, 15, 21 (iii) 12, 72, 31, 32 (iv) 17, 18, 192, 1021

10. (a) Use the extended Euclidean algorithm to express the following as a linear combination.
(i) GCD (144, 89) (ii) GCD (1001, 100001)
(b) Solve each of these congruences using the modular inverses.
(i) $55x \equiv 34 \pmod{89}$ (ii) $89x \equiv 2 \pmod{232}$
(c) Find an inverse of a modulo m for each of these pairs of relatively prime integers.
(i) a = 2, m = 17 (ii) a = 34, m = 89 (iii) a = 144, m = 233 (iv) a = 200, m = 1001

11. (a) Ali and Fatima are cousins. Both are good in Number Theory. They challenge one-another to solve different problems.
(i) Ali gifted Fatima a book entitled "Discrete Mathematics and Its Applications". Suppose that first 9 digits of ISBN-10 of the textbook are *125973128*. How can Fatima find the check digit to validate the originality of the book?
(ii) Ali asks Fatima to show him that 15 is an inverse of 7 modulo 26.
(b)
(i) Fatima has asked Ali to encrypt the message "*exam*" using the RSA system with $n = 11 \cdot 3$ and $e = 3$. translate each letter into integers and write in the form of Cipher text equation.
(ii) But suddenly their cousin Bilal comes, Fatima tells Ali "*XYTU YFQPNSL*". This message is encoded using function *f(p) = (p + 5) mod 26.* Decode the message.
(c)
(i) Fatima asks Ali to list all integers between −100 and 100 that are congruent to −1 modulo 25.
(ii) Bilal asks Ali to determine the remainder of 5^{119} upon division by 59?

12. (a) Use the construction in the proof of the Chinese remainder theorem to find all solutions to the system of congruences.
(i) $x \equiv 1 \pmod 5$, $x \equiv 2 \pmod 6$, and $x \equiv 3 \pmod 7$. (ii) $x \equiv 1 \pmod 2$, $x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$, and $x \equiv 4 \pmod{11}$.
(b) An old man goes to market and a camel step on his basket and crushes the oranges. The camel rider offers to pay for the damages and asks him how many oranges he had brought. He does not remember the exact number, but when he had taken them out 5 at a time, there were 3 oranges left. When he took them 6 at a time, there were also 3 oranges left, when he had taken them out 7 at a time, there was only 1 orange was left and when he had taken them out 11 at a time, there was no orange left. What is the number of oranges he could have had?
(c) A birthday boy receives a certain number of presents. If they are arranged in piles of 4, 3 presents are left over. If arranged in piles of 9, 5 presents are left over. If arranged in piles of 13, 8 presents are left over. What is the smallest number of presents he received?

13. (a)
(i) Encrypt the message "I LOVE DISCRETE MATHEMATICS" by translating the letters into numbers, applying the Caesar Cipher Encryption function and then translating the numbers back into letters.
(ii) Encrypt the message STOP POLLUTION by translating the letters into numbers, applying the given encryption function, and then translating the numbers back into letters.
(i) f (p) = (p + 4) mod 26 (ii) f (p) = (p + 21) mod 26
(b) Decrypt these messages encrypted using the Caesar Cipher.
(i) PLG WZR DVVLJQPHQW (ii) IDVW QXFHV XQLYHUVLWB
(c) Decrypt these messages encrypted using the Shift cipher. f (p) = (p + 10) mod 26.
(i) CEBBOXNOB XYG (ii) LO WI PBSOXN

14. (a) Which memory locations are assigned by the hashing function h(k) = k mod 97 to the records of insurance company customers with these Social Security numbers?
(i) 034567981 (ii) 183211232 (iii) 220195744 (iv) 987255335
(b) Which memory locations are assigned by the hashing function h(k) = k mod 101 to the records of insurance company customers with these Social Security numbers?
(i) 104578690 (ii) 432222187 (iii) 372201919 (iv) 501338753
(c) What sequence of pseudorandom numbers is generated using the linear congruential generator? Perform 5 iterations.
(i) $x_{n+1} = (4x_n + 1) \bmod 7$ with seed $x_0 = 3$? (ii) $x_{n+1} = (7x_n + 2) \bmod 9$ with seed $x_0 = 5$?

15. (a) Determine the check digit for the UPCs that have these initial 11 digits.
    (i) 73232184434 (ii) 63623991346
    (b) Determine whether each of the strings of 12 digits is a valid UPC code.
    (i) 036000291452 (ii) 012345678903
    (c)
    (i) The first nine digits of the ISBN-10 of the European version of the fifth edition of this book are 0-07-119881.What is the check digit for that book?
    (ii)The ISBN-10 of the sixth edition of Elementary Number Theory and Its Applications is 0-321-500Q1-8, where Q is a digit. Find the value of Q.

16. (a) Encrypt the following messages "ATTACK" and "EXAMINATION" using the RSA system with n = 43 · 59 and e = 13, translating each letter into integers and grouping together pairs of integers.
    (b) Encrypt the message "ASSIGNMENT" and "SEMESTER" using the RSA system with n = 13 · 19 and e = 47.
    (c) Use Fermat's little theorem to compute the following.
    (i) 5^{2003} mod 7 (ii) 5^{2003} mod 11 (iii) 5^{2003} mod 13 (iv) 5^{2003} mod 17

17. (a) Ali is very fond of playing Cricket. He has collected tennis ball in a Basket. He's not sure exactly how many he has today, so when he
    aligns them in rows of 7 balls, 2 balls are left;
    aligns them in rows of 17 balls, 3 balls are left;
    aligns them in rows of 19 balls, 5 balls are left.
    He's positive that there are fewer than 700 balls but how many does he have?
    Hint: In this problem, you are supposed to use the following Theorems:
    Chinese Remainder Theorem, The Euclidean Algorithm Lemma, Bézout's Theorem, and Linear congruences.
    (b) After counting the number of balls Ali had from part (a), encrypt the number word using Ceaser Cipher.
    Hint: you have to convert the number in word counting. i.e., Five hundred and twenty-one.
    (c) Use Fermat's' theorem, what is the value of x^{302} (mod 7), where x is the sum of all the digits of the number obtained in part (a).

18. (a) An office building contains 27 floors and has 37 offices on each floor. How many offices are in the building?
    (b) A particular brand of shirt comes in 12 colors, has a male version and a female version, and comes in three sizes for each sex. How many different types of this shirt are made?
    (c) How many different three-letter initials can people have?

19. (a) A basketball team has five starting players. There are 13 girls on the team. In how many ways can the coach select players to start the game? (Assume each player can play every position.)
    (b) 75 people run the Basking Ridge 5k and medals are only awarded to the 1st, 2nd, 3rd, 4th, and 5th place positions. How many different ways can the runners be awarded?
    (c) An amusement park has 27 different rides. If you have enough time to go on 4 rides, how many differen combinations of rides can you take?

20. (a) A wired equivalent privacy (WEP) key for a wireless fidelity (WiFi) network is a string of either 10, 26, or 58 hexadecimal digits. How many different WEP keys are there?
    (b) How many strings are there of four lowercase letters that have the letter x in them?
    (c) How many different three-letter initials with none of the letters repeated can people have?

21. (a) How many functions are there from the set {1, 2, ... , *m*}, where *m* is a positive integer, to the set {0, 1}?
    (b) How many one-to-one functions are there from a set with five elements to sets with five elements?
    (c) Police use photographs of various facial features to help eyewitnesses identify suspects. One basic identification kit contains 15 hairlines, 48 eyes and eyebrows, 24 noses, 34 mouths, and 28 chins and 28 cheeks. Find the total number of different faces.

22. (a) Use a tree diagram to determine the number of subsets of {3, 7, 9, 11, 24} with the property that the sum of the elements in the subset is less than 28.
    (b) How many bit strings of length four do not have two consecutive 1's?

(c) Three officers, a president, a treasurer, and a secretary are to be chosen from among four possible: A, B, C and D. Suppose that A cannot be president and either C or D must be secretary. How many ways can the officers be chosen?

23. (a) Eight members of a school marching band are auditioning for 3 drum major positions. In how many ways can students be chosen to be drum majors?
(b) You must take 6 CS elective courses to meet your graduation requirements at FAST-NUCES. There are 12 CS courses you are interested in. In how many ways can you select your elective Courses?
(c) Nine people in our class want to be on a 5-person basketball team to represent the class. How many different teams can be chosen?

24. (a) A committee of five people is to be chosen from a group of 20 people. How many different ways can a chairperson, assistant chairperson, treasurer, community advisor, and record keeper be chosen?
(b) A relay race has 4 runners who run different legs of the race. There are 16 students on your track team. In how many ways can your coach select students to compete in the race? Assume that the order in which the students run matters.
(c) Your school yearbook has an editor in chief and an assistant editor in chief. The staff of the yearbook ha s 15 students. In how many ways can a student be chosen for these 2 positions?

25. (a) A deli offers 5 different types of meat, 3 types of breads, 4 types of cheeses and 6 condiments. How many different types of sandwiches can be made of 1 meat, 2 bread, 1 cheese, and 3 condiments?
(b) An ice cream shop has a choice of 12 toppings. In how many ways can you have 5 different toppings on your ice cream?
(c) Determine how many different 5 digit postal zip codes are possible for the given configuration with and without repetition. (i) Begins with a 4. (ii) Begins with a 3 or a 1.

26. (a) How many bit strings of length 10 either begin with three 0s or end with two 0s?
(b) How many bit strings of length 5 either begin with 0 or end with two 1s?
(c) For the given configurations, determine how many different computer passwords are possible if repetition is allowed and if repetition is not allowed. (i) 2 letters followed by 4 digits. (ii) 3 odd digits followed by 2 vowels.

27. (a) Show that if there are 30 students in a class, then at least two have last names that begin with the same letter.
(b) Assuming that no one has more than 1,000,000 hairs on the head of any person and that the population of New York City was 8,008,278 in 2010, show there had to be at least nine people in New York City in 2010 with the same number of hairs on their heads.
(c) There are 38 different time periods during which classes at a university can be scheduled. If there are 677 different classes, how many different rooms will be needed?

28. (a) What is the coefficient of:
(i) x^5 in $(1 + x)^{11}$ (ii) a^7b^{17} in $(2a - b)^{24}$
(b)
(i) Write the first six rows of Pascal's Triangle.
(ii) Expand $(3x - 2)^4$ using Pascal's Triangle.
(c) Evaluate:
(i) $(2x + y)^6$ (ii) $(3a - 2b)^7$

29. (a) A class has 20 women and 16 men. In how many ways can you:
(i) put all the students in a row? (ii) put 7 of the students in a row?
(iii) put all the students in a row if all the women are on the left and all the men are on the right?
(b) A baseball coach is determining a batting order for the nine players on the field. If the pitcher does not bat in one of the first four spots, how many different batting orders can the coach make?
(c) A pizza shop offers twelve different toppings. How many different pizzas can you order with no more than three toppings?

30. (a) Mr. Shoaib Raza is teaching Discrete structures this semester and is preparing a final exam. He has a pool of 15 questions on Graph theory, 12 questions on Number theory, 7 questions on Propositional logic, 5 questions on trees, 9 questions on Relations and 10 questions on Set theory. He wishes to put 3 questions on each topic on the exam. Determine the number of different ways the above things can happen.
(b) Suppose that the grades earned by the 97 students appeared in final exam for Discrete Structures class this semester are A, A-, B+, B, B-, C+, C, C-, D and F. At least how many students will earn the exact same grade?

(c) FAST-NUCES cricket team squad consists of 15 players which includes 2 wicket-keepers, 5 bowlers, 3 all-rounders and 5 batsmen. In how many ways, can cricket squad choose a captain, a vice-captain and a wicket-keeper from among themselves?

31. As we have discussed, the practical application of all the topics in the class. Now you are required to submit at least two real world applications of the following topics.

(a) Combination
(b) Permutations
(c) Binomial Theorem
(d) Proof methods
(e) Mathematical Induction